

EXERCISES



PART A – COMPETENCY FOCUSED QUESTIONS

1. Multiple Choice Questions:
Tick (✓) the correct answer:

1.

1			
2	3		
4	5	6	
7	8	9	10

Floyd's Triangle

- Name the type of loop in Java, depicted in the above pattern of Floyd's Triangle.
- (a) Use of two loops (b) Nested loop
(c) Pattern loop (d) Conditional loop
2. In a nested loop, which of the following loop can be used as an outer loop?
(a) while (b) do-while (c) for (d) All of these
3. Which technique is followed in a nested loop structure?
(a) Outer loop iterates under inner loop.
(b) Inner loop iterates under outer loop.
(c) Outer and inner loops iterate simultaneously.
(d) Outer and inner loops iterate alternately.
4. Which of the following statements is not valid for a nested loop?
(a) The break statement can be used to terminate inner as well as outer loop.
(b) The outer loop can be terminated from the block of inner loop.
(c) The inner loop can be terminated from the block of outer loop.
(d) Inner loop repeats the execution a number of times for each iteration of the outer loop.

5. Given is a nested loop construct:

```
for(x = 1; x < 3; x++)
```

```
{
```

```
    for(y = 1; y < x; y++)
```

```
    {
```

```
        .....
```

```
        Statements
```

```
        .....
```

```
    }
```

```
}
```

How many times the above nested loop will iterate?

- (a) 2 (b) 3 (c) 4 (d) 1

6. The labelled break statement used under inner loop will terminate:

- (a) Outer loop (b) Inner loop
(c) Both the loops simultaneously (d) Outer loop followed by inner loop

7. A program snippet using nested loop construct is given below:

```
s = 0;
for(i = 1; i <= 3; i++)
{
    for(j = 1; j < i; j++)
    {
        s += j;
    }
}
```

What will be the final value of 's' when the above program snippet is executed?

- (a) 5 (b) 2 (c) 3 (d) 4

8. Read the following text and choose the correct answer:

A nested loop is a loop used within another loop, i.e., an inner loop within the body of an outer one. The break and continue statements can be used in an outer as well as an inner loop.

Which of the following statements is not valid for a nested loop?

- (a) The break statement can be used to terminate inner as well as outer loop.
(b) The outer loop can be terminated from the block of inner loop.
(c) The inner loop can be terminated from the block of outer loop.
(d) The inner loop repeats the execution a number of times for each iteration of the outer loop.

II. Fill in the blanks:

1. A loop within another loop is called
2. You can use statement to terminate a loop block.
3. is a tag that decides which part of the loop is to be used.
4. Repetition of loop will be completed before loop.
5. statement will repeat a loop for the next iteration after ignoring some of the statements of the loop.

III. Write whether the following statements are True/False:

1. A loop is a repetitive structure.
2. Nesting of loops means restricting the use of inner loop.
3. When a break statement is used, it will terminate the program completely.
4. When the outer loop will complete its iterations, the inner loop starts.
5. Labelled continue statement is used in the inner loop and helps to skip the iterations of outer loop.

IV. Assertion and Reasoning based questions:

Assertion (A): The program written in for loop can be designed using while loop or do-while loop.

Reason (R): These loops can also be useful to perform fixed number of iterations.

Based on the above discussion, choose an appropriate statement from the options given below:

- (a) Both Assertion (A) and Reason (R) are true and Reason (R) is a correct explanation of Assertion (A).
- (b) Both Assertion (A) and Reason (R) are true and Reason (R) is not a correct explanation of Assertion(A).
- (c) Assertion (A) is true and Reason (R) is false.
- (d) Assertion (A) is false and Reason (R) is true.

V. Application based questions:

Your younger brother has tried to write a program using nested loop structure to find the sum of the following series:

$$S = 1 + (1+2) + (1+2+3) + \dots + (1+2+3+\dots+n)$$

Due to some confusion, he could not complete the program and has left some places blank in the statements **Line 1**, **Line 2**, **Line 3** and **Line 4** to be filled with the value/expression. The incomplete program is as shown below:

```
import java.util.*;
class Series
{
    public static void main(String args[ ])
    {
        Scanner in = new Scanner(System.in);
        int i,j,p,n,s = 0;
        System.out.println("Enter the value of n: ");
        n=in.nextInt( );
        for(i = 1; i <= n; i++)
        {
            p = 0;
            for(j = 1; .....; j++)                //Line 1
                p += .....                        //Line 2
            .....                                //Line 3
        }
        System.out.println("Sum of the series:" .....); //Line 4
    }
}
```

Help your brother to complete the above program by answering the following questions:

- (a) What condition will be filled in the blank of **Line 1**?
- (b) What value will be filled in the blank of **Line 2**?
- (c) What statement will be written in the blank of **Line 3**?
- (d) What variable should be written to display the result in the blank of **Line 4**?

VI. Give the output of the following snippets based on nested loops:

```
1. int i,j;
   for (i=0; i<4; i++)
   {
       for (j=i; j>=0; j--)
           System.out.print(j);
```

Output:


```

        System.out.println( );
    }
2. int y,p;
   for (int x=1; x<=3; x++)
   {
       for (y=1; y<=2; y++)
       {
           p = x * y;
           System.out.print(p);
       }
       System.out.println( );
   }

```

Output:

```

3. int a,b;
   for (a=1; a<=2; a++)
   {
       for (b= (64+a); b<=70; b++)
           System.out.print((char) b);
       System.out.println( );
   }

```

Output:

```

4. int x,y;
   for(x=1; x<=5; x++)
   {
       for(y=1; y<x; y++)
       {
           if(x == 4)
               break;
           System.out.print(y);
       }
       System.out.println( );
   }

```

Output:

```

5. int i,j;
   first:
   for (i=10; i>=5; i--)
   {
       for (j= 5; j<=i; j++)
       {
           if (i*j <40)
               continue first;
           System.out.print(j);
       }
       System.out.println( );
   }

```

Output:

VII. Unsolved Java Programs:

1. Write a program to display the Mathematical Table from 5 to 10 for 10 iterations in the given format:

Sample Output: Table of 5

$$5 \times 1 = 5$$

$$5 \times 2 = 10$$

$$5 \times 10 = 50$$

2. Write a program to accept any 20 numbers and display only those numbers which are prime.

[Hint: A number is said to be prime, if it is divisible only by 1 and the number itself.]

3. Write two separate programs to generate the following patterns using iteration (loop) statements:

[ICSE-2015]

(a) *

* #

* # *

* # * #

* # * # *

(b) 5 4 3 2 1

5 4 3 2

5 4 3

5 4

5

4. Write a program to calculate and display the factorials of all the numbers between 'm' and 'n' (where $m < n$, $m > 0$, $n > 0$).

[Hint: factorial of 5 means: $5! = 5 \times 4 \times 3 \times 2 \times 1$]

5. Write a menu driven program to display all the prime and non-prime numbers from 1 to 100.

Enter 1: to display all prime numbers

Enter 2: to display all non-prime numbers

[Hint: A number is said to be prime if it is divisible only by 1 and the number itself.]

6. Write a program to display the factorial of any ten numbers.

[Hint: Factorial of $5! = 5 \times 4 \times 3 \times 2 \times 1$]

Sample Input: 8

Factorial of 8 = 40320

Factorial of 11 = 39916800

and so on.

7. Write a program to input a number. Calculate and display the factorial of each digit.

Sample Input: 365

Factorial of 5 = 120

Factorial of 6 = 720

Factorial of 3 = 6

8. Write a program to input a number and display all its prime factors.

Sample Input: 84

Sample Output: 2, 2, 3, 7

9. Write a Java program to input a number and check whether the number is Unique or not.
A number is said to be a Unique number, if it is a positive integer (without leading zeros) with no duplicate digits.
For example: 123, 234, 456, are Unique numbers.

10. Write a program to input a number (3-digit or more). Arrange the digits of the number in ascending order and display the result.

Sample Input: 8425

Sample Output: 2 4 5 8

11. Write a program to display all the composite numbers from 1 to 100. A number is said to be composite, if it has two or more factors excluding 1 and the number itself.

Sample Input: 6

Sample Output: Factors of 6 are 2 and 3.

Hence, 6 is a composite number.

Few composite numbers are 4, 6, 8, 9, 10, 12, 14, 15, 16, 18, 20, 21,

12. In a school, there are four different sections. In each section, there are 40 students who have appeared for ICSE Examination. Write a program to input percentage marks of each student of each section. Calculate and display the number of students of each section, securing 95% and above in the council examination.

13. Write programs to find and display the sum of the given series:

(a) $1 + \frac{1}{2!} + \frac{1}{3!} + \frac{1}{4!} + \dots + \frac{1}{n!}$

(b) $1 + (1+2) + (1+2+3) + \dots + (1+2+3+\dots+n)$

(c) $1 + (1*2) + (1*2*3) + \dots + (1*2*3*\dots*n)$

(d) $1 + \frac{1}{1+2} + \frac{1}{1+2+3} + \dots + \frac{1}{1+2+3+\dots+n}$

(e) $\frac{1+2}{1*2} + \frac{1+2+3}{1*2*3} + \dots + \frac{1+2+3+\dots+n}{1*2*3*\dots*n}$

14. Write the programs in Java to display the following patterns:

(a) 1

2 1

3 2 1

4 3 2 1

5 4 3 2 1

(c) 5 4 3 2 1

5 4 3 2

5 4 3

5 4

5

(b) 1 2 3 4 5

1 2 3 4

1 2 3

1 2

1

(d) 1 3 5 7 9

1 3 5 7

1 3 5

1 3

1

(r) 5
5 4
5 4 3
5 4 3 2
5 4 3 2 1

(s) 9 9 9 9 9
7 7 7 7 7
5 5 5 5 5
3 3 3 3 3
1 1 1 1 1

(t) 9
9 7
9 7 5
9 7 5 3
9 7 5 3 1

(f) 1 2 3 4 5
2 3 4 5
3 4 5
4 5
5

(h) 9
7 9
5 7 9
3 5 7 9
1 3 5 7 9

(j) 1
2 3
4 5 6
7 8 9 10
11 12 13 14 15

[ICSE-2016]

PART B – SUBJECTIVE QUESTIONS

I. Differentiate between the following:

1. Nested if and nested loop
2. Break and continue
3. Labelled break and Unlabelled break

II. Answer the following questions:

1. What do you mean by a nested loop?
2. In what situation do you need a nested loop?
3. How will you terminate outer loop from the block of the inner loop?
4. What do you mean by labelled break statement? Give an example.
5. Write down the syntax of the nested loop.



Lab Activity

There are two special patterns given below. Write two different programs to display these patterns and finally show the outputs to your teacher.
[Hint: Use two different nested loops to execute these patterns.]

```
*  
* *  
* * *  
* * * *  
* * * * *  
* * * *  
* * *  
* *  
*
```

```
1 2 3 4 5  
1 2 3 4  
1 2 3  
1 2  
1  
1 2  
1 2 3  
1 2 3 4  
1 2 3 4 5
```

```
1 1 1 1 1  
2 2 2 2 2  
3 3 3 3 3  
4 4 4 4 4  
5 5 5 5 5
```

```
5 5 5 5 5  
4 4 4 4 4  
3 3 3 3 3  
2 2 2 2 2  
1 1 1 1 1
```